



Processing Big Data

AWS Academy Data Engineering

Introduction

Processing Big Data

Module objectives

This module prepares you to do the following:

- Compare and select the big data processing framework that best supports your workloads.
- Explain the principles of Apache Hadoop and Amazon EMR, and how they support data processing in AWS.
- Launch, configure, and manage an Amazon EMR cluster to support big data processing.

Module overview

Presentation sections

- Big data processing concepts
- Apache Hadoop
- Apache Spark
- Amazon EMR
- Managing your Amazon EMR clusters
- Apache Hudi

Labs

- Processing Logs by Using Amazon EMR
- Updating Dynamic Data in Place

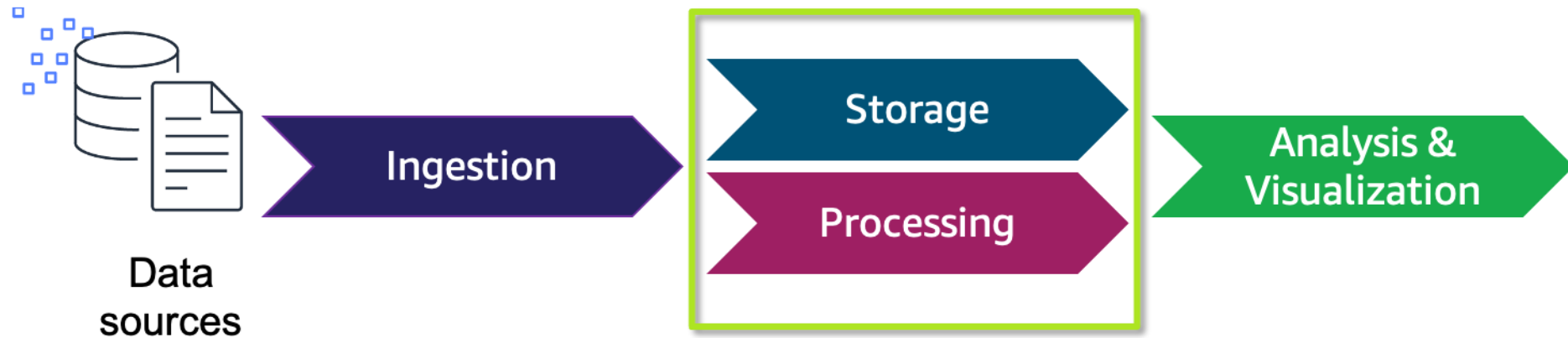
Knowledge checks

- Online knowledge check
- Sample exam question

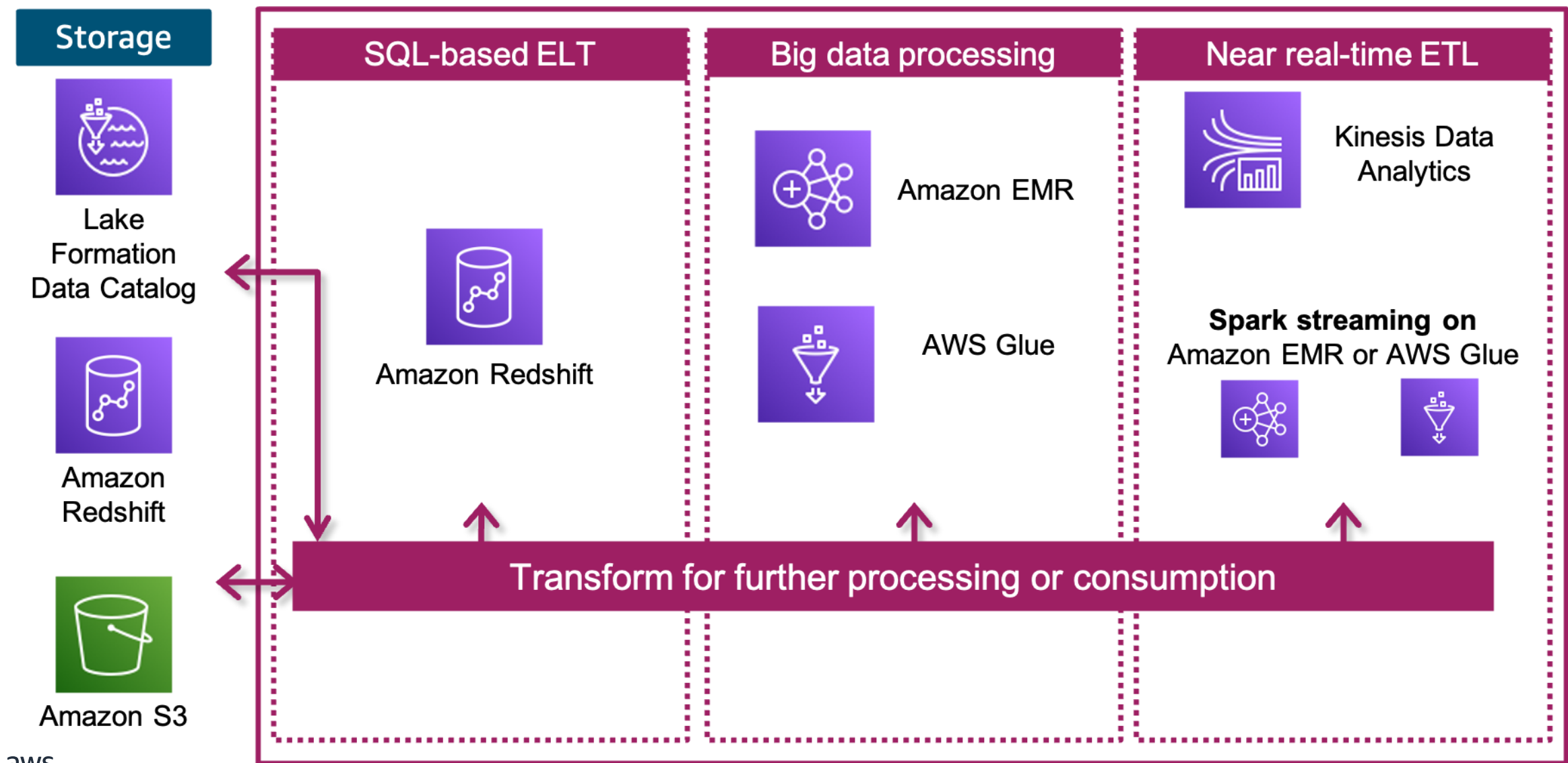
Big data processing concepts

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Big data processing and the iterative data pipeline



Big data processing in the modern architecture pipeline



Types of data processing

Batch data processing

- Infrequently accessed (cold) data querying
- Processes input data in batches at varying intervals
- Tolerates structured and unstructured data
- Capable of deep analysis of big datasets
- Examples: Amazon EMR, Apache Hadoop

Streaming data processing

- Frequently accessed (hot) data querying
- Processes data sequentially and incrementally in near real time
- Capable of processing less predictable data on a massive scale
- Enables analysis of continually generated data
- Examples: Amazon Kinesis Data Streams, Apache Spark Streaming

Frameworks that support batch and stream processing

Batch Processing	Stream Processing	
Apache Spark	Amazon Kinesis	
Apache Hadoop MapReduce	Apache Spark Streaming	AWS Lambda
Apache Hive	Apache Hive	Apache Flink
Apache Pig	Apache Pig	Apache Storm

Key takeaways: Big data processing concepts



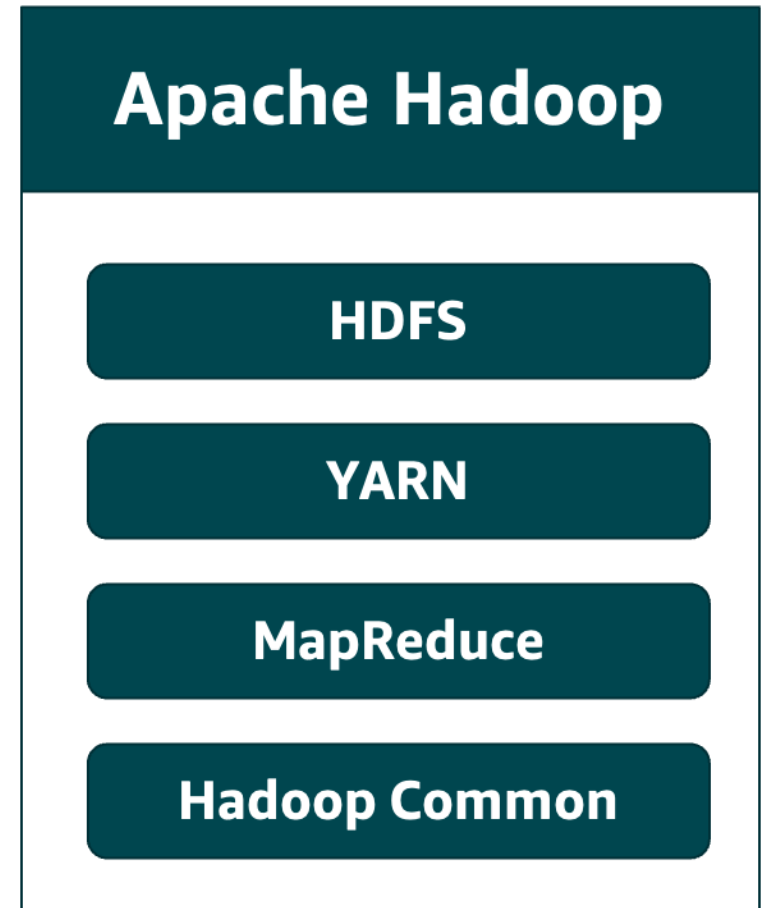
- Big data processing is generally divided into two categories, batch and streaming.
- Batch data typically involves cold data, with analytics workloads that involve longer processing times.
- Streaming data involves many data sources providing data that must be processed sequentially and incrementally.
- Batch processing and stream processing each benefit from specialized big data processing frameworks.

Apache Hadoop

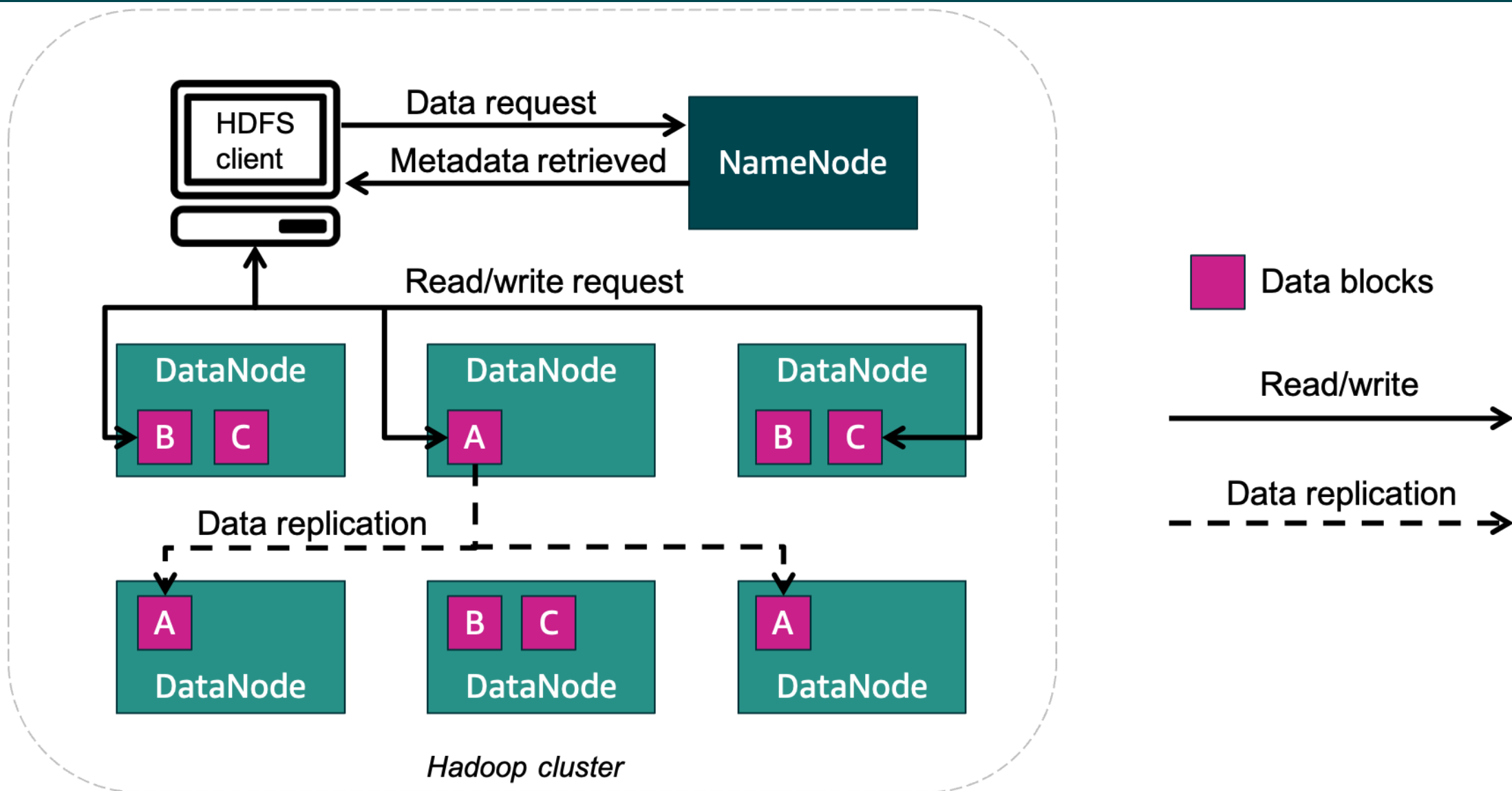
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Apache Hadoop characteristics

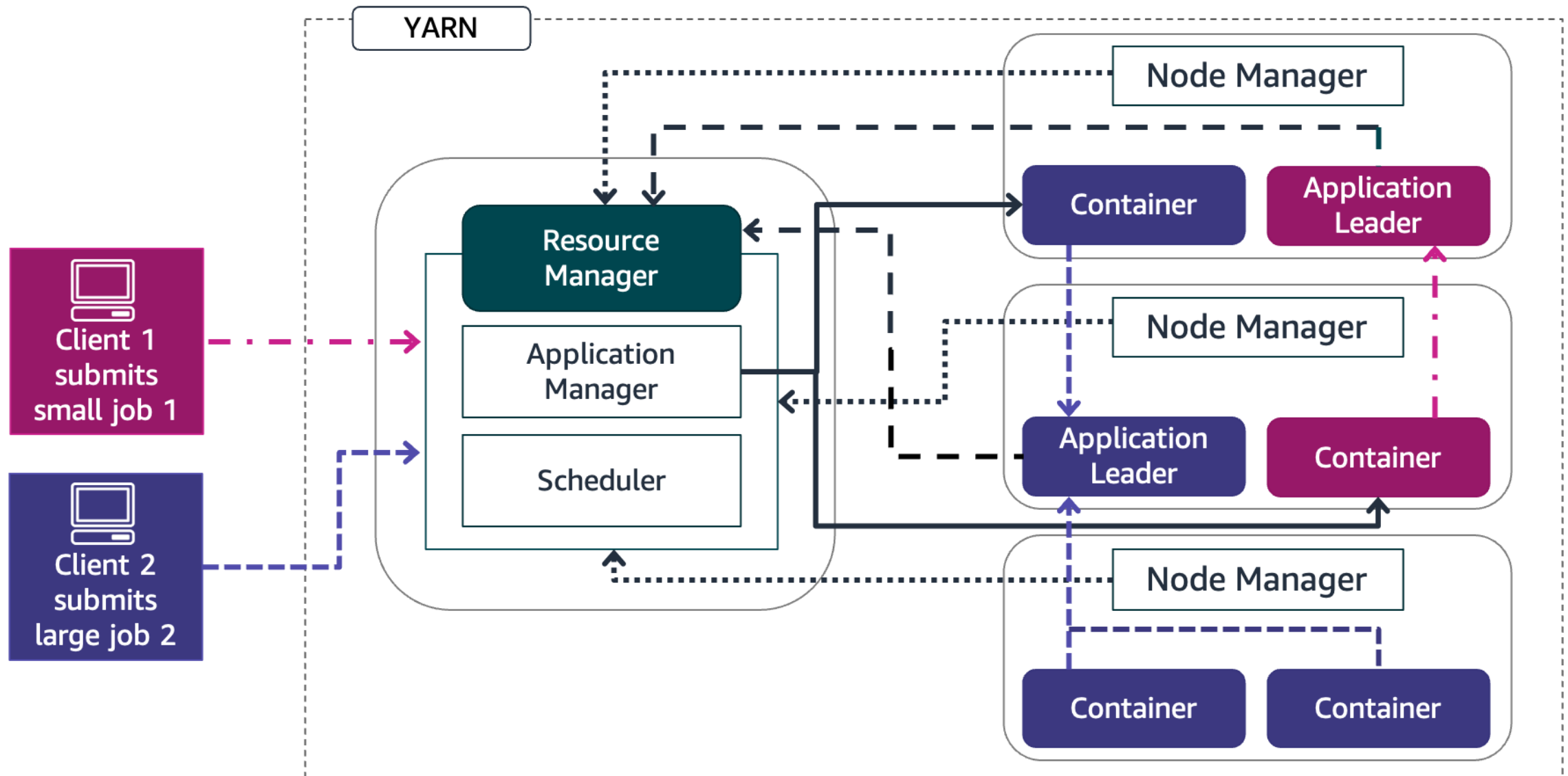
- Is an open-source, distributed processing framework
- Enables distributed storage and processing for large amounts of data
- Maps tasks to nodes within clusters of servers
- Hadoop components: Hadoop Distributed File System (HDFS), YARN, MapReduce, Hadoop Common
- Clusters consist of main nodes and worker nodes



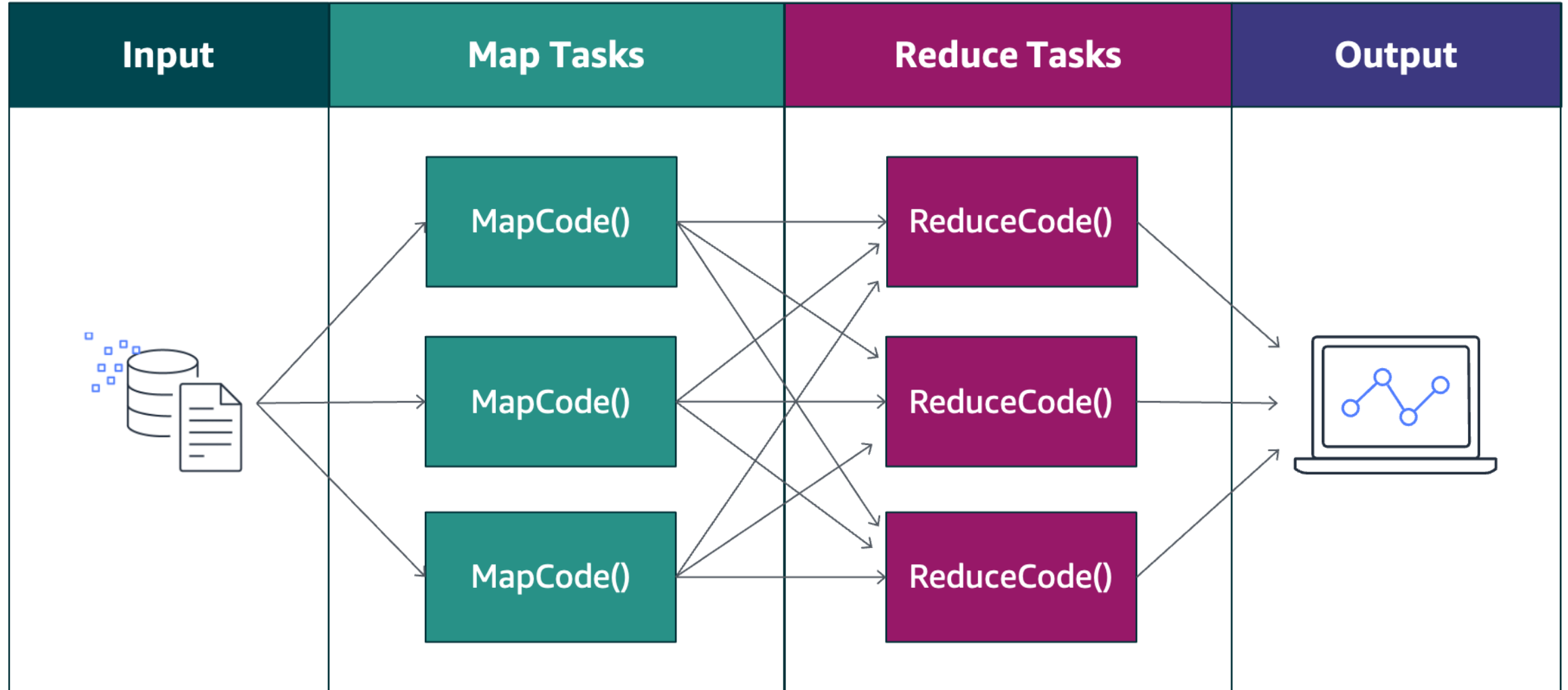
Hadoop Distributed File System (HDFS)



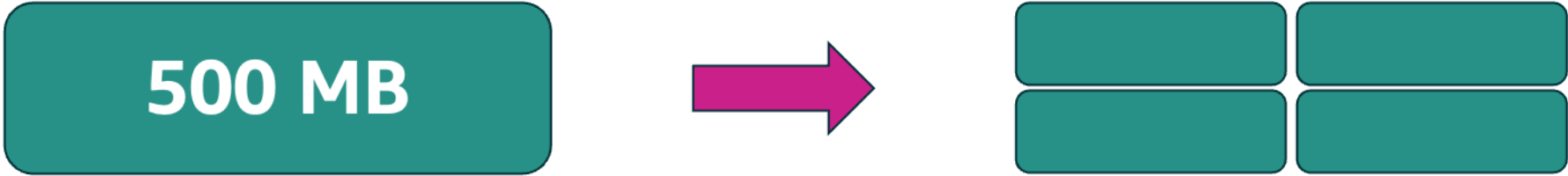
Yet Another Resource Negotiator (YARN)



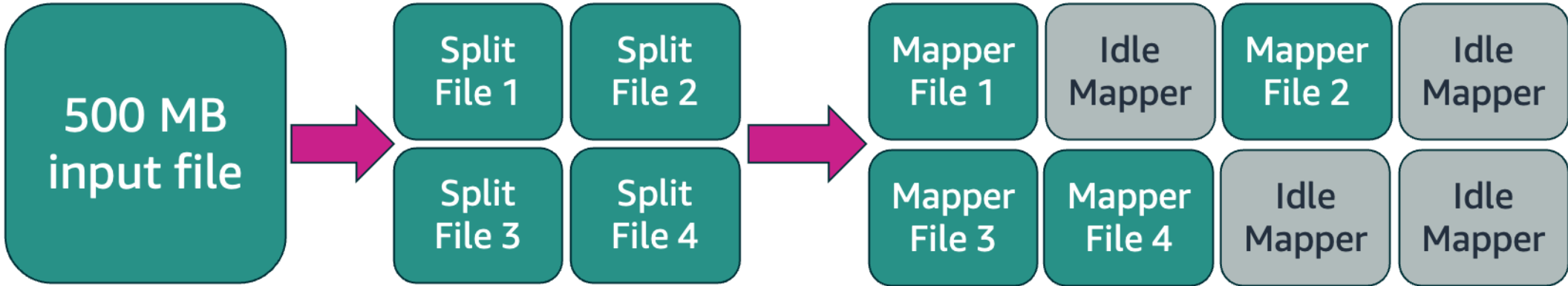
Hadoop MapReduce



Processing data with Hadoop MapReduce



A single file from Amazon S3 is split into four parallel HTTP requests



The Hadoop default split results in four files, using four of the eight available mappers

Common Hadoop frameworks to process big data

Apache Flink	Apache Hive	Presto	Apache Pig
• Streaming data flow engine • Uses APIs that are optimized for distributed streaming and batch processing • Provides the ability to perform transformations on data sources • API is categorized into DataSets and DataStreams	• Open-source, SQL-like data warehouse solution • Helps you avoid needing to write complex MapReduce programs in lower level computing languages • Integrates with AWS services such as Amazon S3 and Amazon DynamoDB	• Open-source, in-memory SQL query engine • Emphasizes faster querying for interactive queries • Operates in-memory and is based on its own engine	• Textual data flow language • Performs analysis of large datasets using parallel processing • Supports automatic install when an Amazon EMR cluster is launched • Supports interactive development

Key takeaways: Apache Hadoop



- Hadoop includes a distributed storage system, HDFS.
- When you store data in HDFS, Hadoop splits the data into smaller data blocks.
- MapReduce processes large datasets with a parallel, distributed algorithm on a cluster.

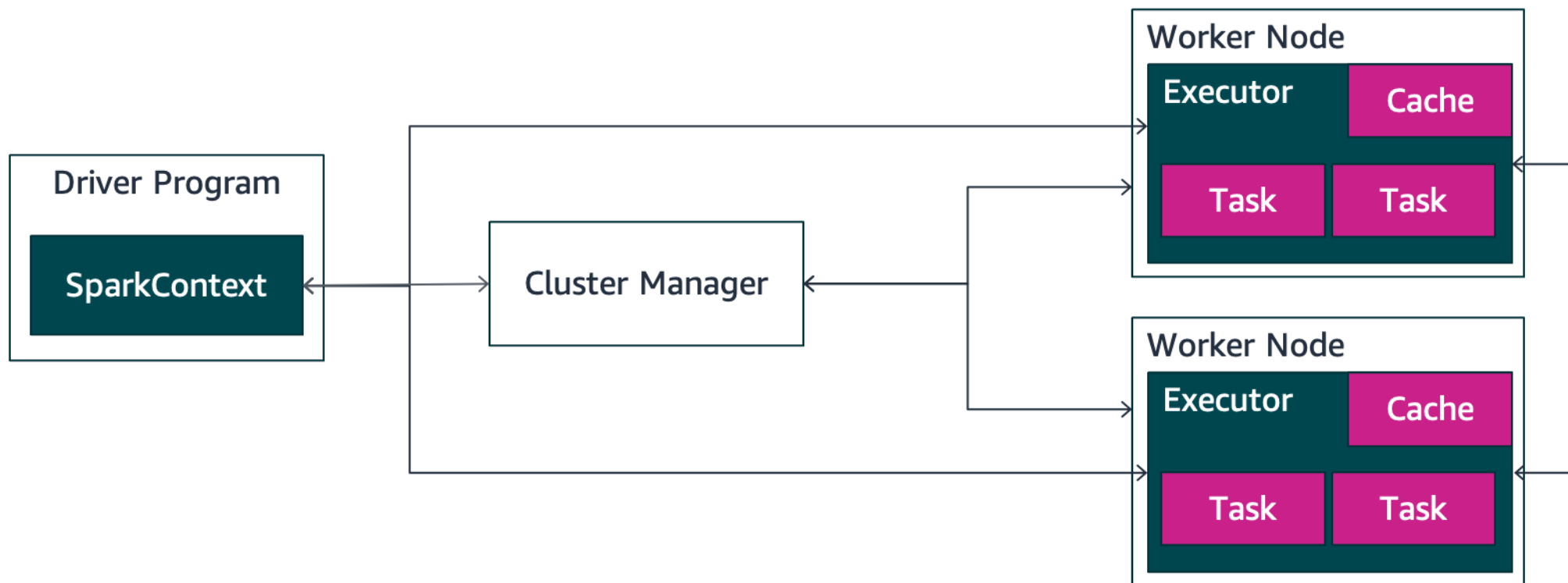
Apache Spark

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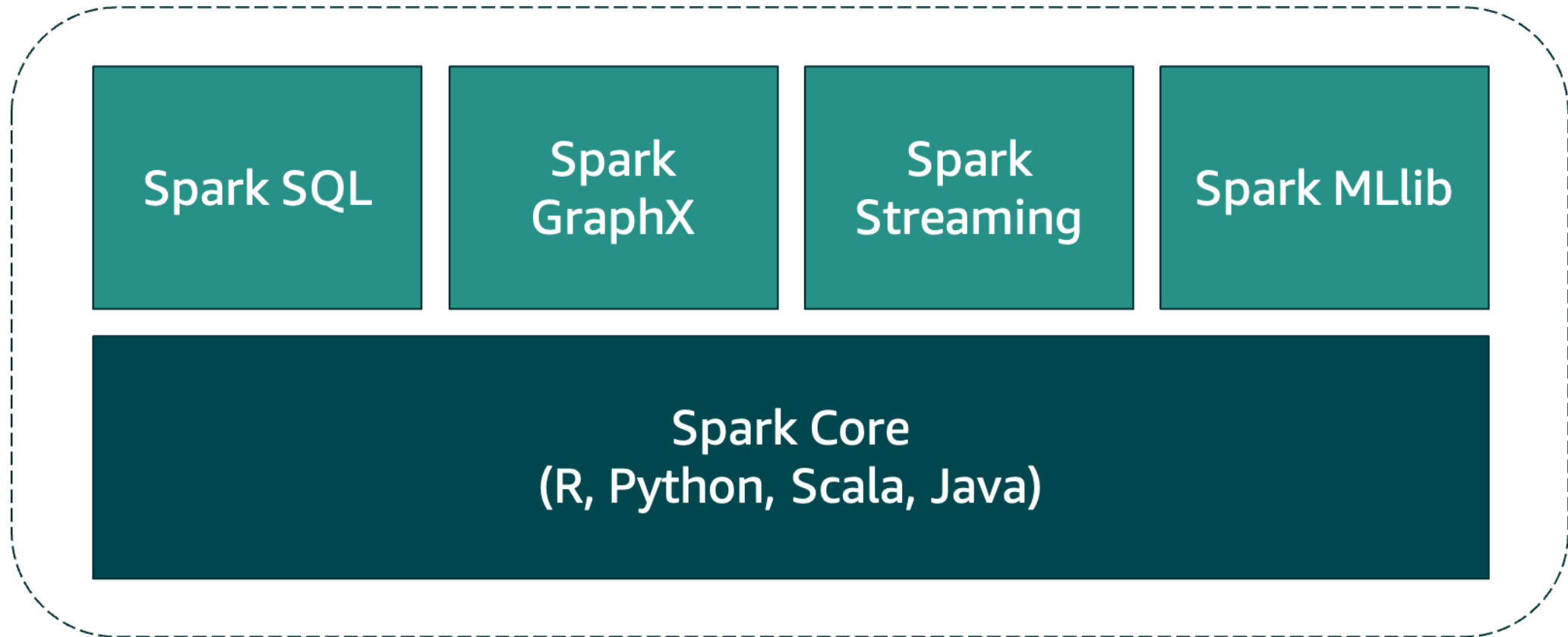
Apache Spark characteristics

- Is an open-source, distributed processing framework
- Uses in-memory caching and optimized query processing
- Supports code reuse across multiple workloads
- Clusters consist of leader and worker nodes

Spark clusters



Spark components



Key takeaways: Apache Spark



- Apache Spark performs processing in-memory, reduces the number of steps in a job, and reuses data across multiple parallel operations.
- Spark reuses data by using an in-memory cache to speed up ML algorithms.

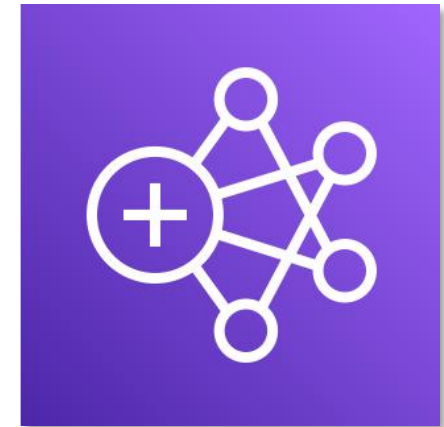
Amazon EMR

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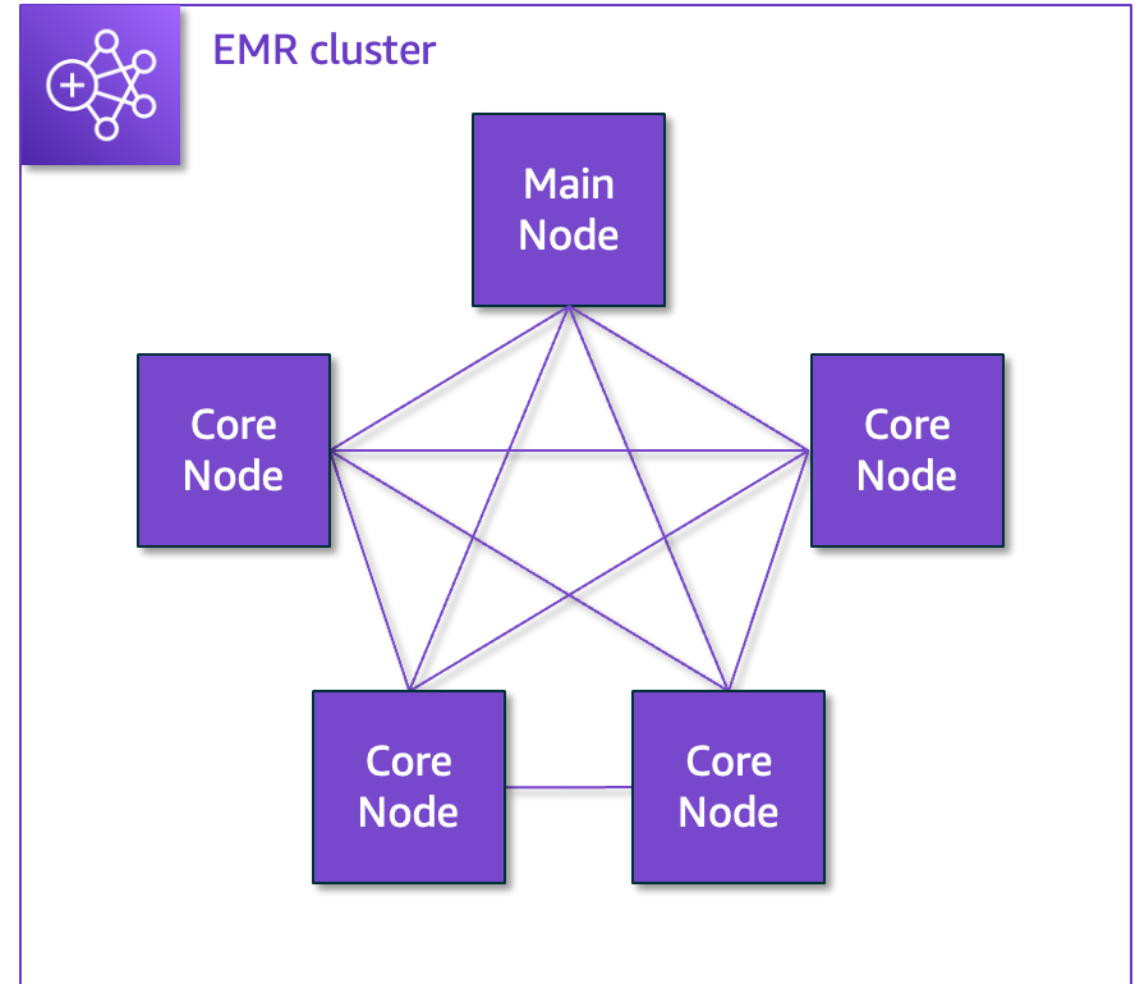
Amazon EMR characteristics

- Managed cluster platform
- Big data solution for petabyte-scale data processing, interactive analytics, and machine learning
- Processes data for analytics and BI workloads using big data frameworks
- Transform and move large amounts of data into and out of AWS data stores



Clusters and nodes

- The central component of Amazon EMR is the *cluster*.
- Each instance in the cluster is a *node*.
- The role that each node serves is the *node type*.
- Amazon EMR uses three node types: main, core, and task.

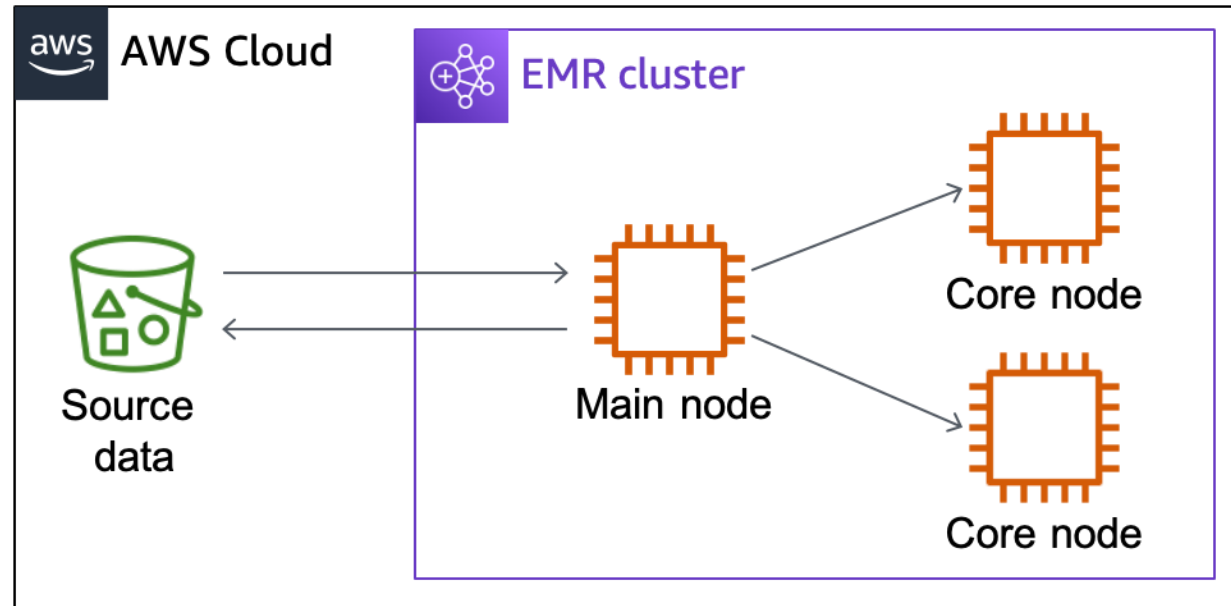


Amazon EMR service architecture

Storage	HDFS, EMR File System (EMRFS), local file system
Cluster resource management	YARN
Data processing frameworks	Hadoop MapReduce, Apache Spark
Applications and programs	Apache Spark, Apache Hive, Apache Flink, Apache Hadoop

The Amazon EMR architecture
consists of multiple layers

Processing data in Amazon EMR



Key takeaways: Amazon EMR



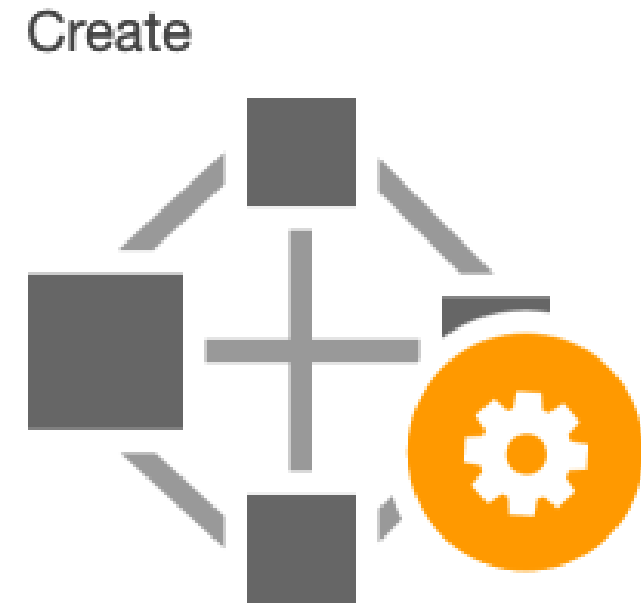
- You can use Amazon EMR to perform automated installations of common big data projects.
- The Amazon EMR service architecture consists of four distinct layers:
 - Storage layer
 - Cluster resource management layer
 - Data processing frameworks layer
 - Applications and programs layer

Managing your Amazon EMR clusters

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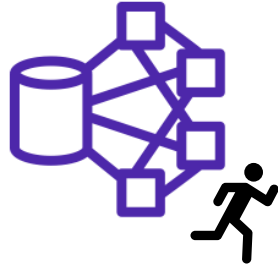
Launching and configuring Amazon EMR clusters

- Three methods are available to launch EMR clusters:
 - Interactive mode
 - Command line mode
 - API mode



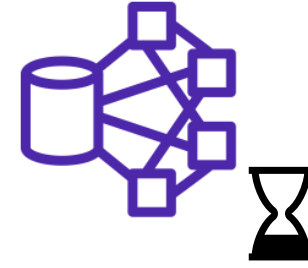
Cluster characteristics

Amazon EMR clusters are characterized in one of two ways:



Long-running clusters

- Persistent clusters
- Interactive jobs submission
- Persistent data (until shutdown)
- Large dataset processing

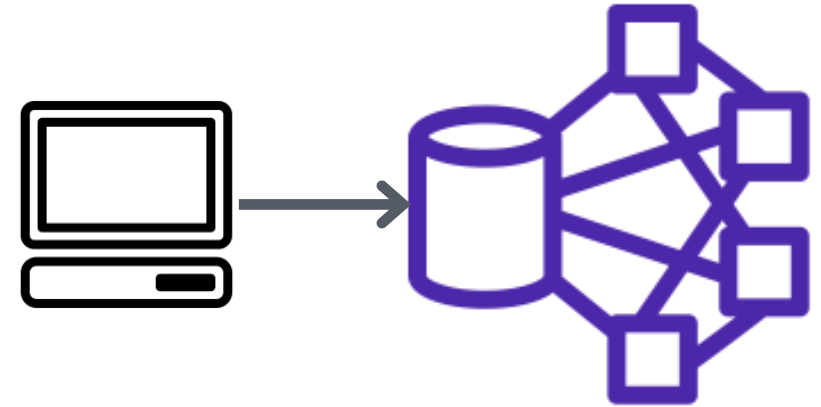


Transient clusters

- Shut down after data is processed and stored
- Typically read code and data from Amazon S3 at startup
- Don't persist HDFS data after termination

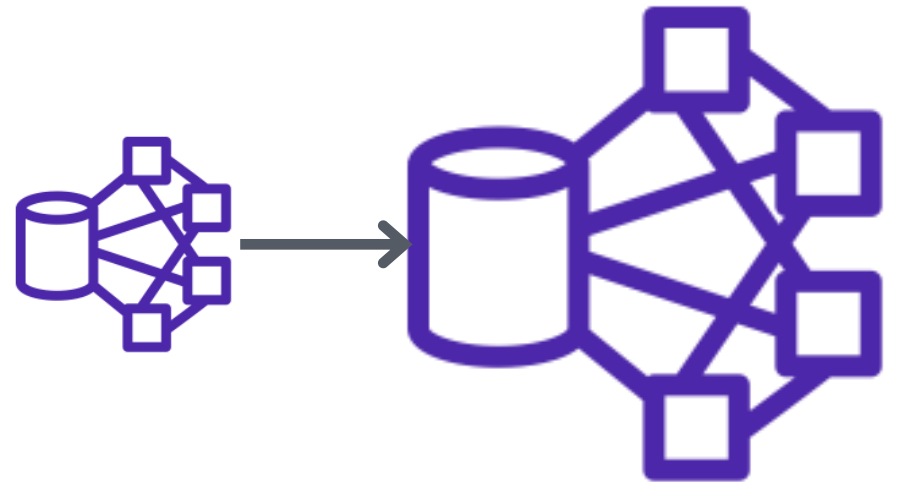
Connecting to your cluster

- External connections are made to the main node Amazon EC2 instance.
 - The main node exposes the public DNS name for connections.
 - Amazon EMR creates security group rules for the main node.
 - Amazon EMR creates separate security group rules for the core and task nodes.
- Clusters must be running for connections to be made.



Scaling your cluster resources

- Scaling is accomplished automatically or manually.
- Two options are available for automatic scaling:
 - Amazon EMR managed scaling
 - Custom automatic scaling policy



Key takeaways: Managing your Amazon EMR clusters



- Three methods are available to launch EMR clusters: interactive, command line, and API.
- EMR clusters are characterized as either long running or transient, based on their usage.
- External connections to EMR clusters can only be made through the main node.

Lab: Processing Logs by Using Amazon EMR

Lab introduction: Processing Logs by Using Amazon EMR



- In this lab, you will use Amazon EMR to process logs that are stored in an S3 bucket.
- To process the data in Amazon EMR, you will use Apache Hive.
- Open your lab environment to start the lab and find additional details about the tasks that you will perform during this lab.

Debrief: Processing Logs by Using Amazon EMR

- In Hive, what is the difference between an external table and a table that isn't defined as external?
- What happens when you run a SELECT query in Hive?
- Are AWS Glue and Athena comparable to Hive? How are they similar or different?

Apache Hudi

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Apache Hudi characteristics

- Is an open-source data management framework
- Provides record-level insert, update, upsert, and delete capabilities
- Integrates with Apache Spark, Apache Hive, and Presto
- Hudi DeltaStreamer utility provides the ability to create or update Hudi datasets
- Organizes a dataset into a partitioned directory structure under a base path, similar to a Hive table

Key Hudi concepts

Hudi dataset storage types

Copy on Write (CoW)

- Stores data in columnar format (Parquet)
- Each update creates a new version of files during the write process
- Default storage type

Merge on Read (MoR)

- Stores data in a combination of columnar format (Parquet) and row-based (Avro) formats
- Updates are logged to row-based delta files and compacted as needed

Hudi view options

Read-optimized view

Incremental view

Real-time view

Key takeaways: Apache Hudi



- Apache Hudi provides the ability to ingest and update data in near real time.
- Hudi maintains metadata of the actions that are performed to ensure those actions are both atomic and consistent.

Lab: Updating Dynamic Data in Place

Lab introduction: Updating Dynamic Data in Place



- In this lab, you will use Amazon S3, Amazon Athena, AWS Glue, and Apache Hudi to address the challenge of accommodating a dynamically changing schema.
- You will use these services to facilitate efficient in-place data updates and run queries to get data in near real time.
- Open your lab environment to start the lab and find additional details about the tasks that you will perform during this lab.

Debrief: Updating Dynamic Data in Place

- How did you capture streaming data and store it in Amazon S3 in this lab?
- What is the purpose of the schema evolution capability in Apache Hudi?
- After you changed the schema of the data that Kinesis was publishing, how did you update the Athena query?

Module wrap-up

Processing Big Data

Module summary

This module prepared you to do the following:

- Compare and select the big data processing framework that best supports your workloads.
- Explain the principles of Apache Hadoop and Amazon EMR, and how they support data processing in AWS.
- Launch, configure, and manage an Amazon EMR cluster to support big data processing.

Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

Sample exam question

A data engineer has deployed an Amazon EMR cluster to support their ML workload. Additional cluster configuration is needed, but numerous attempts to use SSH to connect to the active EC2 instance, which is acting as the main node of the cluster, have failed.

What could cause this connectivity issue?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- **Amazon EMR cluster**
- **SSH**
- **Connect to the active EC2 instance**
- **Main node of the cluster**

Sample exam question: Response choices

A data engineer has deployed an **Amazon EMR cluster** to support their ML workload. Additional cluster configuration is needed, but numerous attempts to use **SSH to connect to the active EC2 instance**, which is acting as the **main node of the cluster**, have failed.

What could cause this connectivity issue?

Choice	Response
A	Connections to an EMR cluster must be made directly to the desired core node.
B	Amazon EMR does not support SSH connections to clusters.
C	The ElasticMapReduce-main security group needs an inbound rule that allows HTTPS access.
D	The ElasticMapReduce-main security group needs an inbound rule that allows SSH access.

Sample exam question: Answer

The correct answer is D.

Choice	Response
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D	The ElasticMapReduce-main security group needs an inbound rule that allows SSH access.
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Thank you



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